

Bias/variance in linear regression models

The in-sample prediction error under squared loss is

$$\text{ErrF} = \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m \{ \tilde{y}_i - \hat{f}(x_i) \}^2 \right] = \mathbb{E} [\text{MSE}_{\text{test}}]$$

Because of the results shown in the slides, we have

$$\text{ErrF} = \sigma^2 + \underbrace{\frac{1}{m} \sum_{i=1}^m \mathbb{E} \left[\hat{f}(x_i) - f(x_i) \right]^2}_{\text{Bias}^2} + \underbrace{\frac{1}{m} \sum_{i=1}^m \text{var}(\hat{f}(x_i))}_{\text{Variance}}$$

If the estimate is a linear model, $\hat{f}(x) = x^T \hat{\beta}$, then...

$$\begin{aligned} \text{Bias}(x_i) &= \mathbb{E} [\hat{f}(x_i) - f(x_i)] = \mathbb{E} [x_i^T \hat{\beta} - f(x_i)] = \\ &= \mathbb{E} (x_i^T \hat{\beta}) - f(x_i) = x_i^T (X^T X)^{-1} X^T \mathbb{E} [y] - f(x_i). \end{aligned}$$

In our case, $\mathbb{E} [y] = f$, $f = (f(x_1), \dots, f(x_m))^T$.

$$\text{Bias}(x_i) = x_i^T (X^T X)^{-1} X^T f - f(x_i). \quad (\text{Data mining assumptions})$$

Note that $f(x)$ is a generic function. However, in the lucky situation in which
a.k.a. non-linear

it holds that $f(x_i) = x_i^T \beta$, then $f = X\beta$ and

$$\text{Bias}(x_i) = x_i^T \cancel{(X^T X)^{-1} X^T} X \beta - x_i^T \beta = x_i^T \beta - x_i^T \beta = 0. \quad (\text{Model I assumptions}).$$

As for the variance, we get:

$$\begin{aligned} \text{Variance} &= \frac{1}{m} \sum_{i=1}^m \text{var} \{ \hat{f}(x_i) \} = \frac{1}{m} \sum_{i=1}^m \text{var} \{ x_i^T \hat{\beta} \} = \frac{1}{m} \sum_{i=1}^m \text{var} \{ x_i^T (X^T X)^{-1} X^T y \} \\ &= \frac{1}{m} \sum_{i=1}^m x_i^T (X^T X)^{-1} X^T \underbrace{\text{var}(y)}_{\sigma^2 I_m} (x_i^T (X^T X)^{-1} X^T)^T = \end{aligned}$$

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n x_i^T (x_i^T x_i)^{-1} x_i^T x_i (x_i^T x_i)^{-1} x_i = \frac{\sigma^2}{n} \sum_{i=1}^n x_i^T (x_i^T x_i)^{-1} x_i.$$

↓
symmetric

Now, note that $H = X(X^T X)^{-1} X^T$ and $\text{tr}(H) = \sum_{i=1}^n [H]_{ii} = \sum_{i=1}^n x_i^T (X^T X)^{-1} x_i.$

$$\text{Variance} = \frac{\sigma^2}{n} \text{tr}(H) = \sigma^2 \frac{p}{n}$$

Optimism

The optimism is defined as

$$\text{Opt} = \mathbb{E}[\text{MSE}_{\text{test}}] - \mathbb{E}[\text{MSE}_{\text{train}}]$$

" $\mathbb{E}_{\text{MF}}$

Hence, the optimism is defined as

$$\begin{aligned} \text{Opt} &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \{\tilde{y}_i - \hat{f}(x_i)\}^2\right] - \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \{y_i - \hat{f}(x_i)\}^2\right] \\ &= \frac{1}{n} \sum_{i=1}^n \left(\mathbb{E}[\{y_i^2 - \hat{f}(x_i)\}^2] - \mathbb{E}[\{y_i - \hat{f}(x_i)\}^2] \right) \end{aligned}$$

Let us isolate a specific term in the above summation ($\mathbb{E}[\text{MSE}_{\text{train}}]$)

$$\begin{aligned} \mathbb{E}[\{y_i - \hat{f}(x_i)\}^2] &= \mathbb{E}[\underbrace{z_i^2}_{\substack{\text{"} z_i^2 \text{"}}} = \text{var}(z_i) + \mathbb{E}[z_i]^2 \\ &\quad \text{properties of the variance} \\ &= \text{var}[y_i - \hat{f}(x_i)] + \mathbb{E}[y_i - \hat{f}(x_i)]^2 \\ &\quad \text{can expand, using properties of the variance.} \quad \text{"} (f(x) - \mathbb{E}[\hat{f}(x_i)])^2 \text{"} \\ &\quad \text{" Bias}^2(x_i) \end{aligned}$$

$$\begin{aligned}
 \mathbb{E}[\{y_i - \hat{f}(x_i)\}^2] &= \text{var}(y_i) + \text{var}(\hat{f}(x_i)) - 2\text{cov}(y_i; \hat{f}(x_i)) + \text{Bias}^2(x_i) \\
 &= \sigma^2 + \text{var}(\hat{f}(x_i)) - 2\text{cov}(y_i; \hat{f}(x_i)) + \text{Bias}^2(x_i)
 \end{aligned}$$

Following similar steps, for each term of $\mathbb{E}[\text{MSE}_{\text{test}}]$

$$\mathbb{E}[\{\tilde{y}_i - \hat{f}(x_i)\}^2] = \text{var}(\tilde{y}_i) + \text{var}(\hat{f}(x_i)) - 2\text{cov}(\tilde{y}_i; \hat{f}(x_i)) + \mathbb{E}[\tilde{y}_i - \hat{f}(x_i)]^2$$

!! Because $\hat{f}$ is not based on $\tilde{y}_i$.

Crucially, now we exploit that $y_i \stackrel{d}{=} \tilde{y}_i \Rightarrow \text{var}(y_i) = \text{var}(\tilde{y}_i) = \sigma^2$, and that $\mathbb{E}[y_i] = \mathbb{E}[\tilde{y}_i] = f(x_i)$.

$$= \sigma^2 + \text{var}(\hat{f}(x_i)) + \text{Bias}^2(x_i)$$

Let us put these pieces together:

$$\begin{aligned}
 \text{Opt} &= \frac{1}{n} \sum_{i=1}^n \left(\mathbb{E}[\{y_i - \hat{f}(x_i)\}^2] - \mathbb{E}[\{y_i - f(x_i)\}^2] \right) \\
 &= \frac{1}{n} \sum_{i=1}^n \left(\cancel{\sigma^2} + \text{var}(\hat{f}(x_i)) + \text{Bias}^2(x_i) - \cancel{\sigma^2} - \text{var}(\hat{f}(x_i)) + 2\text{cov}(y_i, \hat{f}(x_i)) - \cancel{\text{Bias}^2(x_i)} \right) \\
 &= \frac{2}{n} \sum_{i=1}^n \text{cov}(y_i; \hat{f}(x_i)) = \frac{2}{n} \text{tr} \{ \text{cov}(y, \hat{f}) \}.
 \end{aligned}$$

If the estimate is a linear model, $\hat{f}(x) = x^T \hat{\beta}$, then...

$$\begin{aligned}
 \text{Opt}_{\text{ols}} &= \frac{2}{n} \sum_{i=1}^n \text{cov}(y_i; \hat{f}(x_i)) = \frac{2}{n} \text{tr} \{ \text{cov}(y, \hat{f}) \} \\
 &= \frac{2}{n} \text{tr} \{ \text{cov}(y, Hy) \} = \frac{2}{n} \text{tr} \{ \text{cov}(y, y) H^T \} \\
 &= \frac{2}{n} \text{tr}(\sigma^2 H^T) = 2 \frac{\sigma^2}{n} \text{tr}(H^T) = 2 \frac{\sigma^2}{n} \text{tr}(H) = 2\sigma^2 \frac{p}{n}.
 \end{aligned}$$

LOO-CV for linear models

Let us first define some notation:

$$e_{-i}^2 = (y_i - \hat{y}_{-i})^2 = (y_i - x_i^T \hat{\beta}_{(-i)})^2 = (X_{(-i)}^T X_{(-i)})^{-1} X_{(-i)}^T y_{(-i)} \quad (\text{LOO-CV residuals})$$

$$X_{(-i)}^T X_{(-i)} = X^T X - x_i x_i^T; \quad X_{(-i)}^T y_{(-i)} = X^T y - x_i y_i.$$

$$h_i = x_i^T (X^T X)^{-1} x_i = [X(X^T X)^{-1} X^T]_{ii} = [H]_{ii}$$

Then, note that

$$\hat{\beta}_{(-i)} = [X^T X - x_i x_i^T]^{-1} [X^T y - x_i y_i]$$

Sherman-Morrison

$$= \left[(X^T X)^{-1} + \frac{1}{1 - h_i} (X^T X)^{-1} x_i x_i^T (X^T X)^{-1} \right] [X^T y - x_i y_i]$$

$$= \underbrace{(X^T X)^{-1} X^T y}_{\hat{\beta}} - (X^T X)^{-1} x_i y_i + \frac{1}{1 - h_i} (X^T X)^{-1} x_i x_i^T (X^T X)^{-1} [X^T y - x_i y_i]$$

$$= \hat{\beta} - (X^T X)^{-1} x_i y_i + \frac{1}{1 - h_i} (X^T X)^{-1} x_i x_i^T \underbrace{(X^T X)^{-1} X^T y}_{\hat{\beta}} - \frac{1}{1 - h_i} (X^T X)^{-1} x_i x_i^T \underbrace{(X^T X)^{-1} x_i y_i}_{h_i}$$

$$= \hat{\beta} - \frac{(X^T X)^{-1} x_i}{1 - h_i} \left[\underbrace{(1 - h_i) y_i - x_i^T \hat{\beta} + h_i y_i}_{\substack{y_i - x_i^T \hat{\beta} \\ e_i} - \cancel{h_i y_i} + \cancel{h_i y_i}} \right]$$

$$= \hat{\beta} - \frac{(X^T X)^{-1} x_i}{1 - h_i} \cdot e_i = \hat{\beta}_{(-i)}.$$

Hence, we get

$$e_{-i} = y_i - x_i^T \hat{\beta}_{-i} = \underbrace{y_i - x_i^T \hat{\beta}}_{e_i} + \frac{\overbrace{x_i^T (X^T X)^{-1} x_i}^{h_i}}{(1 - h_i)} \cdot e_i$$

$$e_{-i}^1 = e_i + \frac{h_i}{1-h_i} \cdot e_i = \frac{e_i - \cancel{e_i h_i} + \cancel{e_i h_i}}{1-h_i} = \frac{e_i}{1-h_i}$$

In other words:

$$y_i - \hat{y}_{-i} = \frac{y_i - \hat{y}_i}{1-h_i}.$$

Ridge estimator

Let us define the penalized loss $e(\beta)$ as

$$e(\beta) = (y - X\beta)^T(y - X\beta) + \lambda \beta^T \beta.$$

The ridge estimator is

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta \in \mathbb{R}^p} e(\beta).$$

Let us consider the gradient of $e(\beta)$, which is:

$$\begin{aligned} e^*(\beta) &= \frac{\partial}{\partial \beta} e(\beta) = \frac{\partial}{\partial \beta} (y - X\beta)^T(y - X\beta) + \frac{\partial}{\partial \beta} \lambda \beta^T \beta \\ &= -2X^T(y - X\beta) + 2\lambda \beta \end{aligned}$$

To identify the stationary points, we set

$$\begin{aligned} -2X^T(y - X\beta) + 2\lambda \beta &= 0 &\iff X^T(y - X\beta) - \lambda \beta &= 0 \\ &&\iff X^T y &= X^T X \beta + \lambda \beta \\ &&\iff X^T y &= (X^T X + \lambda I_p) \beta \end{aligned}$$

$$\hat{\beta} = \underbrace{(X^T X + \lambda I_p)^{-1}}_{\downarrow} X^T y.$$

Now, we let

$\downarrow$ This matrix is always invertible (see later).

$$e^{**}(\beta) = \frac{\partial}{\partial \beta^T} e^*(\beta) = \frac{\partial^2}{\partial \beta \partial \beta^T} e(\beta) = 2X^T X + 2\lambda I_p = 2 \underbrace{\left[X^T X + \lambda I_p \right]}_W.$$

The matrix W is positive definite (hence $\hat{\beta}$ is a minimum) because its eigenvalues that is $(d_1^2 + \lambda, \dots, d_p^2 + \lambda)$, are strictly positive for $\lambda > 0$.

Variance of PCR and ridge (Unit C, missing details)

Let us recall that we are assuming that

$$y_i^* = f(x_i^*) + \varepsilon_i$$

with y_i^* and x_i^* being the "original" data, so that

$$y_i = y_i^* - \bar{y}^*, \text{ and } x_{ij} = \frac{x_{ij}^* - \bar{x}_j^*}{s_j}.$$

In matrix notation, we have that

$$y = (I_m - \Pi_m) y^*, \quad [\Pi]_{ij} = \frac{1}{n}$$

Moreover, we have that $X = UDV^T$ (singular value decomposition) and $Z = XV = UD$.

PCR estimator

Let us denote with $\hat{\gamma} = (Z^T Z)^{-1} Z^T y$. Hence, we get the following estimate

$$\begin{aligned} \hat{\beta}_{\text{PCR}} &= V \hat{\gamma} = V (Z^T Z)^{-1} Z^T y \\ &= V (Z^T Z)^{-1} \underbrace{Z^T (I_m - \Pi_m)}_{\substack{= \\ Z^T}} y^* \end{aligned}$$

The columns of Z have 0 mean, therefore the centering operation does not have an impact.

$$= V (Z^T Z)^{-1} Z^T y^*$$

Hence,

$$\begin{aligned} \text{var}(\hat{\beta}_{\text{PCR}}) &= \text{var}(V (Z^T Z)^{-1} Z^T y^*) = \sigma^2 V (Z^T Z)^{-1} Z^T [V (Z^T Z)^{-1} Z^T]^T \\ &= \sigma^2 V (Z^T Z)^{-1} \cancel{Z^T Z} (Z^T Z)^{-1} V^T \\ &= \sigma^2 V (Z^T Z)^{-1} V^T = \sigma^2 V (D^T D)^{-1} V^T \end{aligned}$$

$$\text{var}(\hat{\beta}_{\text{pr}}) = \sigma^2 V(D^T D)^{-1} V^T = \sigma^2 \sum_{\delta=1}^k \frac{1}{d_{\delta}^2} \tilde{V}_{\delta} \tilde{V}_{\delta}^T.$$

Variance of ridge regression

The ridge estimator can be written as:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I_p)^{-1} X^T y =$$

$$= (X^T X + \lambda I_p)^{-1} X^T \underbrace{(I_m - \Pi_m)}_{X^T, \text{ because } X \text{ is centered}} y^*$$

$$= (X^T X + \lambda I_p)^{-1} X^T y^*.$$

Hence, the variance is

$$\text{var}(\hat{\beta}_{\text{ridge}}) = \text{var}((X^T X + \lambda I_p)^{-1} X^T y^*) =$$

$$= \sigma^2 (X^T X + \lambda I_p)^{-1} X^T [(X^T X + \lambda I_p)^{-1} X^T]^T$$

$$= \sigma^2 \underbrace{(X^T X + \lambda I_p)^{-1} X^T X}_{V \Delta^2 V^T} \underbrace{(X^T X + \lambda I_p)^{-1}}_{V V^T}$$

$$= \sigma^2 (V \Delta^2 V^T + \lambda I_p)^{-1} V \Delta^2 V^T (V \Delta^2 V^T + \lambda I_p)^{-1}$$

$$= \sigma^2 [V(\Delta^2 + \lambda I_p)V^T]^{-1} V \Delta^2 V^T [V(\Delta^2 + \lambda I_p)V^T]^{-1}$$

$$= \sigma^2 V(\Delta^2 + \lambda I_p)^{-1} \cancel{V^T V} \Delta^2 \cancel{V^T V} (\Delta^2 + \lambda I_p)^{-1} V^T$$

$$= \sigma^2 V(\Delta^2 + \lambda I_p)^{-2} \Delta^2 V^T$$

$$= \sigma^2 \sum_{\delta=1}^p \tilde{V}_{\delta} \frac{d_{\delta}^2}{(d_{\delta}^2 + \lambda)^2} \tilde{V}_{\delta}^T.$$

Effective degrees of freedom of ridge

Suppose $y_i = f(x_i) + \varepsilon_i$, then the ridge predictions can be expressed as

$$\begin{aligned}\hat{y} &= H_{\text{ridge}} (y - \bar{y}) + \bar{y} \\ &= H_{\text{ridge}} (I_m - \Pi_m) y + \Pi_m y\end{aligned}$$

$$= [H_{\text{ridge}} (I_m - \Pi_m) + \Pi_m] y$$

Now, note that

$$\begin{aligned}H_{\text{ridge}} (I_m - \Pi_m) &= X (X^T X + \lambda I_p)^{-1} \underbrace{X^T (I_m - \Pi_m)}_{X^T} = H_{\text{ridge}} \\ &= [H_{\text{ridge}} + \Pi_m] y\end{aligned}$$

$$\begin{aligned}df_{\text{ridge}} &= \frac{1}{\sigma^2} \sum_{i=1}^m \text{cov}(y_i, x_i^T \hat{\beta}_{\text{ridge}}) = \frac{1}{\sigma^2} \text{tr}(\text{cov}(y, (H_{\text{ridge}} + \Pi_m) y)) \\ &= \frac{1}{\sigma^2} \text{tr}[\text{cov}(y, y) (H_{\text{ridge}} + \Pi_m)^T] = \cancel{\frac{\sigma^2}{\sigma^2}} \cdot \text{tr}(H_{\text{ridge}} + \Pi_m) = \\ &= \text{tr}(H_{\text{ridge}}) + \text{tr}(\underbrace{\Pi_m}_{1}) = \text{tr}(H_{\text{ridge}}) + 1.\end{aligned}$$

Now, see that

$$\begin{aligned}\text{tr}(H_{\text{ridge}}) &= \text{tr}(U D (D^T D + \lambda I_p)^{-1} D^T U^T) = \text{tr}(\cancel{U^T U} D (D^T D + \lambda I_p)^{-1} D^T) \\ &\quad \text{see slides for these steps} \\ &= \text{tr}(D (D^T D + \lambda I_p)^{-1} D^T) \\ &= \sum_{\delta=1}^P \frac{d_{\delta}^2}{d_{\delta}^2 + \lambda}.\end{aligned}$$

Local linear regression

The local linear regression estimator is obtained as

$$\hat{f}(x) = \hat{\beta}_1 + \hat{\beta}_2 \cancel{(x-x)} = \hat{\beta}_1,$$

where

$$(\hat{\beta}_1, \hat{\beta}_2) = \underset{(\beta_1, \beta_2)}{\operatorname{argmin}} \sum_{i=1}^m w_i(x) \{y_i - \beta_1 - \beta_2(x - x_i)\}^2.$$

It can be shown (weighted least squares), that

$$\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2) = (X^T W_x X)^{-1} X^T W_x y,$$

where the rows of X are $x_i = (1, x_i - x)$ and $W_x = \operatorname{diag}\{w_1(x), \dots, w_m(x)\}$.

Thus, we get:

$$\bullet \quad X^T W_x X = \begin{bmatrix} \sum_{i=1}^m w_i(x) & \sum_{i=1}^m w_i(x)(x_i - x) \\ \sum_{i=1}^m w_i(x)(x_i - x) & \sum_{i=1}^m w_i(x)(x_i - x)^2 \end{bmatrix} = m \begin{bmatrix} a_0(x) & a_1(x) \\ a_1(x) & a_2(x) \end{bmatrix}$$

$$\bullet \quad X^T W_x y = \begin{bmatrix} \sum_{i=1}^m w_i(x) y_i \\ \sum_{i=1}^m w_i(x)(x_i - x) y_i \end{bmatrix}$$

$$\Rightarrow \left(\frac{1}{m} X^T W_x X \right)^{-1} = \begin{bmatrix} a_2(x) & -a_1(x) \\ -a_1(x) & a_0(x) \end{bmatrix} \frac{1}{a_2(x)a_0(x) - a_1^2(x)}$$

$$\Rightarrow \hat{\beta}_1 = \left[\left(\frac{1}{m} X^T W_x X \right)^{-1} \left(\frac{1}{m} X^T W_x y \right) \right]_1$$

Just the intercept

$$= \frac{1}{a_2(x)a_0(x) - a_1^2(x)} \cdot a_2(x) \left(\frac{1}{n} \sum_{i=1}^n w_i(x) y_i \right) - a_1(x) \left(\frac{1}{n} \sum_{i=1}^n w_i(x) (x_i - x) y_i \right)$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{w_i(x) \{ a_2(x) - a_1(x)(x_i - x) \} y_i}{a_2(x)a_0(x) - a_1^2(x)}$$

Leave one-out cross-validation (general theorem)

Suppose $\hat{f}(x) = \sum_{i=1}^m s_i(x) y_i$, with $\sum_{i=1}^m s_i(x) = 1$. For most linear smoothers, the following property holds:

$$\hat{y}_{-i} = \frac{1}{1 - s_i(x)} \sum_{j \neq i} s_j(x) y_j. \quad (1)$$

Then, it holds

$$\begin{aligned} \frac{1}{1 - s_i(x)} \sum_{j \neq i} s_j(x) y_j &= \frac{1}{1 - s_i(x)} \left[\sum_{j \neq i} s_j(x) y_j + s_i(x) y_i - s_i(x) y_i \right] = \\ &= \frac{1}{1 - s_i(x)} \left[\underbrace{\sum_{i=1}^m s_i(x) y_i}_{\hat{y}_i} - s_i(x) y_i \right] = \\ &= \frac{1}{1 - s_i(x)} (\hat{y}_i - s_i(x) y_i) \\ \hat{y}_{-i} &= \frac{\hat{y}_i}{1 - s_i(x)} - \frac{s_i(x)}{1 - s_i(x)} y_i \end{aligned}$$

Now, as a consequence of the above, we can write:

$$\begin{aligned} y_i - \hat{y}_{-i} &= y_i + \frac{s_i(x)}{1 - s_i(x)} y_i - \frac{1}{1 - s_i(x)} \hat{y}_i \\ &= \frac{y_i - \cancel{s_i(x) y_i} + \cancel{s_i(x) y_i} - \hat{y}_i}{1 - s_i(x)} \\ &= \frac{y_i - \hat{y}_i}{1 - s_i(x)}. \quad (2) \end{aligned}$$

Remark: these steps are reversible. Hence, $(1) \Leftrightarrow (2)$, because the proof works in both directions.